Modeling and Robust Control of Heterogeneous Vehicle Platoons on Curved Roads Subject to Disturbances and Delays

Liwei Xu, Weichao Zhuang, Guodong Yin, *Member, IEEE*, Chentong Bian, and Huawei Wu

***Abstract*—This paper presents an integrated platoon control framework for heterogeneous vehicles on curved roads with varying slopes, aerodynamic drag, and wireless communication delays. First, a linear heterogeneous vehicle platoon model with decoupled longitudinal and lateral dynamics is built based on nonlinear feedback linearization and arc-length parametric transformation of an arbitrary curved road. The external disturbances are also considered, including the up-down road slopes, wind and equivalent time-delays caused by the wireless communication delay and discrete data in onboard sensors. Second, to achieve stable platoon control on a curved road, the platoon uses the state-feedback control method in the longitudinal direction while each vehicle tracks the centerline of the curved road to realize lateral control. Longitudinally, an H_∞ platoon controller is designed based on the Lyapunov-Krasovskii functional to ensure the inner-vehicle and string stability of the platoon, which suffers from parameter uncertainty, and external disturbances. Additionally, the $\mathcal{L}_2$ string stability is defined to guarantee that the perturbations do not grow unbounded as they propagate through the platoon. Simulation results indicate that the proposed robust controller achieves proper platoon intervehicle spacing in both the longitudinal and lateral directions.**

***Index Terms*—Heterogeneous vehicle platoon, curved road, string stability, robust control.**

NOMENCLATURE

δ_i	road slope angle.
$\hat{\vartheta}_i^\theta$	ideal tangential angle of the ith vehicle on the point P_{idea}.
$\hat{\vartheta}_i^Y$	ideal lateral position of the ith vehicle on the point P_{idea}.
ω_i	yaw rate of the ith vehicle.
ρ_s	minimal sample frequency in sensor.
ρ_w	minimal sample frequency in wireless communication.
τ_d	equivalent delay caused by data dropout.
τ_t	information delay in the wireless communication.
θ_i	heading angle of the ith vehicle.
ς_d	total delay in wireless communication.
ς_t^s	data delay generated by discrete data in the sensor.
ς_t^w	data delay generated by discrete data in the wireless communication.
a_0	acceleration of leading vehicles.
C_γ	lateral equivalent aerodynamic resistance coefficient.
C_ε	longitudinal equivalent aerodynamic resistance coefficient.
f	rolling resistance coefficient.
F_i^γ	generalized lateral force of the ith vehicle.
F_i^ε	generalized longitudinal force of the ith vehicle.
g	gravitational acceleration.
I_i^z	inertia moment of the ith vehicle.
n_d	maximum number of packet dropouts in the wireless communication.
P_{idea}	ideal location of the ith vehicle on the curved road.
S_d	ideal intervehicle spacing.
S_i	arc length of vehicles i traveled.
T_d	maximum number of sampling periods in wireless communication.
$v_{f,\gamma}$	lateral wind speed.
$v_{f,\varepsilon}$	longitudinal wind speed.
M_i	mass of the ith vehicle.
v_i^γ	lateral speed of the ith vehicle in the vehicle coordinate system $\boldsymbol{\varepsilon}-\mathbf{o}-\boldsymbol{\gamma}$.
v_i^ε	longitudinal speed of the ith vehicle in the vehicle coordinate system $\boldsymbol{\varepsilon}-\mathbf{o}-\boldsymbol{\gamma}$.
X_i	longitudinal coordinate of the ith vehicle in the inertial frame $\mathbf{X}-\mathbf{O}-\mathbf{Y}$.
Y_i	lateral coordinate of the ith vehicle in the inertial frame $\mathbf{X}-\mathbf{O}-\mathbf{Y}$.

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I. INTRODUCTION

THE VEHICLE platooning that coordinates a group of vehicles driving together has received significant research attention due to its potential in reducing traffic congestion. To realize safe and compact vehicle driving, all of the cars in the platoon are required to run at a harmonized speed under

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prespecified intervehicle spacing [1]. Compared to human-drivn vehicles, a closer headway between vehicles could reduce the aerodynamic drag, and consequently, improve the traffic and vehicle efficiency [2].

Preliminary platooning experiments could go back as far as the Advanced Transportation Technology (PATH) project in California of USA, in the 1990s, in which the modeling, design, and control of platoons were investigated [3]. Since then, various pilot programs were conducted to improve the performance of platooning around the world. In 2011, the Netherlands held a competition, i.e., the Grand Cooperative Driving Challenge (GCDC) to demonstrate and promote the applications of vehicle platooning.

Researchers around the world have investigated vehicle platoons from the perspectives of spacing policy [4], [5], information topology [6], [7], [8], delays in communication and vehicle dynamics [9], [10], homogeneity and heterogeneity [11], [12], centralized and decentralized control [13], [14], and so on. All of these studies focus on improving the string stability of the platoon, i.e., preventing the errors that are magnified along the platoon. Bernardo *et al.* [15] investigated the string stability of platoon with time-varying heterogeneous communication delays and derived sufficient conditions to ensure string stability using a distributed consensus strategy and the Lyapunov-Razumikhin theorem. Ploeg *et al.* [16] studied the string stability of the vehicle platoon under the speed fluctuation of the leading vehicle based on the H_∞ control method, and presented an $\mathcal{L}_2$ string criterion for the vehicle platoon. Zheng *et al.* [17] designed a distributed H_∞ controller to maintain the string stability of a large-scale homogeneous platoon system subject to external disturbances. To guarantee the stability and robustness of a large platoon of vehicles, Hao and Barooah [18] established the scaling laws of the H_∞ norms of the formation to measure the external disturbances. In addition, when the information among the vehicles that is transmitted by wireless communication and radar is discrete in the actual platoon system, the string stability of the platoon under the discrete information approach is investigated in [19] and [20] by sampled-data based platooning control.

Although the pieces of the literature that were referred to above have researched the string stability of a platoon that is suffering communication delays, parameter uncertainty, extraneous interference and discrete properties of the information, few take account all of the factors. Additionally, most of the existing research focuses on the control of a platoon that is driven on a straight lane. However, a vehicle platoon operated on a curved road is rarely studied, in which the platoon dynamics, both longitudinal and lateral, should be considered. Therefore, this paper focuses on the modeling and stable control of heterogeneous vehicle platoons on a curved road subject to external disturbances and communication delays.

The main contributions of this paper are threefold. First, the dynamics of the heterogeneous vehicle platoon system on a curved road is decomposed into longitudinal and lateral models by adopting feedback control linearization and arc-length transformation. The longitudinal platoon model is used to ensure the string stability of the vehicle platoon along a curved road, while the lateral model is established to achieve the road tracking performance of the platoon. Second, in the platoon's longitudinal direction, an H_∞ based robust controller is designed by using the Lyapunov-Krasovskii theorem to reduce the system's conservatism. In the lateral direction, each vehicle adopts a robust tracking controller based on the H_∞ method. Third, to ensure the string stability of the platoon, the $\mathcal{L}_2$ stability is defined for the platoon subjective to the external disturbances and communication delays.

The remainder of this paper is organized as follows. The dynamics modeling of a vehicle platoon on the curved road is proposed in Section II. Section III designs the platoon longitudinal robust controller and the vehicles lateral tracking controller. Section IV introduces the $\mathcal{L}_2$ string stability of the vehicle platoon. Section V validates and analyzes the proposed robust controller by simulations. Section VI concludes the paper.

II. Platoon System Modeling on a Curved Road

As shown in Fig. 1, suppose that a heterogeneous vehicle platoon that is composed of $n+1$ vehicles with different masses, including one leader indexed by 0, and n following vehicles indexed from 1 to n, runs along a curved road. All of the following vehicles in the platoon are hypothesized to be the battery electric vehicles (BEVs), which are driven by in-wheel motors (IWMs). Since the drive motor is directly connected to the wheel, the inertial lag caused by the transmission system in traditional automobiles is not considered in the following vehicle dynamics. In addition, the predecessor-leader following (PLF) communication topology, which has a tremendous advantage in ensuring the string stability of the platoon [1], is adopted to satisfy the information transfer among the vehicles. In this topology, each follower not only picks up the states (the speed and location) of the leading vehicle by wireless communication but also detects the states of its predecessor using radar or lidar.

A. Vehicle Dynamics Model

To depict the vehicle's dynamic behaviour on a curved route, the conventional six degrees-of-freedom double-track vehicle model is employed in this paper. For the ith following vehicle, the dynamics model can be expressed as

$$\begin{cases} \dot{X}_i = v_i^\varepsilon \cos\theta_i - v_i^\gamma \sin\theta_i \\ \dot{Y}_i = v_i^\varepsilon \sin\theta_i + v_i^\gamma \cos\theta_i \\ \theta_i = \omega_i \\ \dot{v}_i^\varepsilon = \omega_i v_i^\gamma + \dfrac{1}{M_i}\left(F_i^\varepsilon - C_\varepsilon v_{f,\varepsilon}^2 \right. \\ \qquad \left. - M_i g f \cos(\delta_i) - M_i g \sin(\delta_i)\right) \\ \dot{v}_i^\gamma = -\omega_i v_i^\varepsilon + \dfrac{1}{M_i}\left(F_i^\gamma - C_\gamma v_{f,\gamma}^2\right) \\ \dot{\omega}_i = \dfrac{T_i^z}{I_i^z} \end{cases} \tag{1}$$

where v_i^ε and v_i^γ are the longitudinal and lateral speeds of the vehicle in the vehicle coordinates $\boldsymbol{\varepsilon} - \mathbf{o} - \boldsymbol{\gamma}$; X_i and Y_i are the longitudinal and lateral coordinates of the vehicle in the

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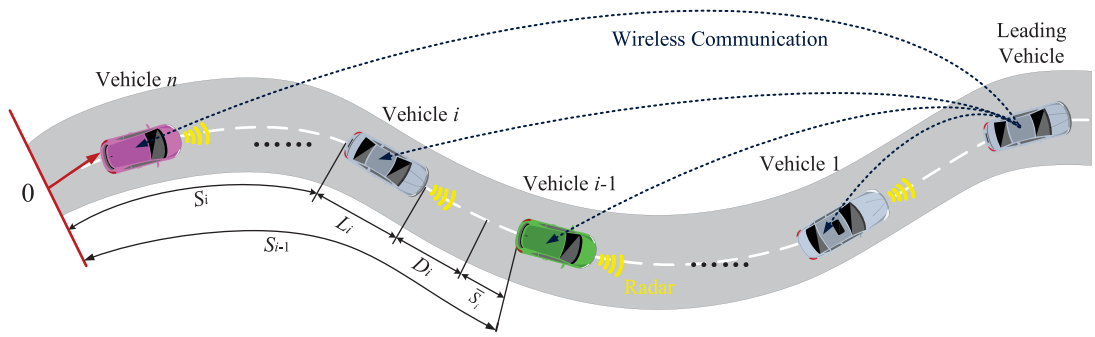

Fig. 1. The predecessor-leader-following-type vehicle platoon on the curved road.

inertial frame $\mathbf{X}-\mathbf{O}-\mathbf{Y}$; M_i and I_i^z are the vehicle mass and the moment of inertia; ω_i is the yaw rate; θ_i is the heading angle; F_i^ε and F_i^γ are the generalized longitudinal and lateral forces; T_i^z is the generalized external moment; C_ε and C_γ are the equivalent aerodynamic resistance coefficients; $v_{f,\varepsilon}$ and $v_{f,\gamma}$ are the wind speeds; g is the gravitational acceleration; f is the rolling resistance coefficient; and δ_i is the road slope angle.

Note that, since the vehicle platoon in this paper is heterogeneous, the parameters M_i and I_i^z are different for each of the following vehicles. Additionally, because the distribution and intensity of the instantaneous airflow are difficult to obtain in reality, the longitudinal and lateral wind speeds are treated as external disturbances in this paper. In addition, considering the position error of the positioning system, the real-time precise road slope angle is hardly achieved. Therefore, it is also a disturbance in this paper. Moreover, to address these interferences conveniently, we assume that these interferences meet the energy bounded condition, i.e., $\int_0^\infty \|v_{f,\varepsilon}(t)\|^2 dt < \infty$, $\int_0^\infty \|v_{f,\gamma}(t)\|^2 dt < \infty$ and $\int_0^\infty \|\delta_i(t)\|^2 dt < \infty$, where $\|\bullet\|$ is the 2-norm.

To facilitate the establishment of the vehicle platoon system, the nonlinear vehicle model (1) must be linearized first. Denote $x_i = [X_i, Y_i, \theta_i, v_i^\varepsilon, v_i^\gamma, \omega_i]^T$ and $u_i = [F_i^\varepsilon, F_i^\gamma, T_i^z]^T$ as the state and input of vehicle system (1) respectively. According to [21], if the output of the vehicle system is $\hbar_i = [X_i, Y_i, \theta_i]^T$, then the vehicle model (1) can be linearized and decoupled into three linear subsystems, which are expressed by

$$\begin{cases} \dot{\vartheta}_{1,i}^X = \vartheta_{2,i}^X \\ \dot{\vartheta}_{2,i}^X = \nu_i^X + \mathcal{G}_i^X w_i^X \end{cases} \tag{2a}$$

$$\begin{cases} \dot{\vartheta}_{1,i}^Y = \vartheta_{2,i}^Y \\ \dot{\vartheta}_{2,i}^Y = \nu_i^Y + \mathcal{G}_i^Y w_i^Y \end{cases} \tag{2b}$$

$$\begin{cases} \dot{\vartheta}_{1,i}^\theta = \vartheta_{2,i}^\theta \\ \dot{\vartheta}_{2,i}^\theta = \nu_i^\theta \end{cases} \tag{2c}$$

where $\vartheta_i = [\vartheta_{1,i}^X, \vartheta_{2,i}^X, \vartheta_{1,i}^Y, \vartheta_{2,i}^Y, \vartheta_{1,i}^\theta, \vartheta_{2,i}^\theta]^T$ is the new state; $\nu_i = [\nu_i^X, \nu_i^Y, \nu_i^\theta]^T$ is the new input; w_i^X and w_i^Y are the disturbances, and $\mathcal{G}_i^X$ and $\mathcal{G}_i^Y$ are the their coefficient matrices, which can be written as

$$\mathcal{G}_i^X = [-fg, -g, -C_\varepsilon/M_i, C_\gamma/M_i] \tag{3a}$$

$$\mathcal{G}_i^Y = [-g, -fg, -C_\varepsilon/M_i, -C_\gamma/M_i] \tag{3b}$$

$$\begin{aligned} w_i^X = {} & [\cos(\delta_i)\cos(x_{3,i}), \sin(\delta_i)\cos(x_{3,i}), \\ & v_{f,\varepsilon}^2 \cos(x_{3,i}), v_{f,\gamma}^2 \sin(x_{3,i})]^T \end{aligned} \tag{3c}$$

$$\begin{aligned} w_i^Y = {} & [\sin(\delta_i)\sin(x_{3,i}), \sin(x_{3,i})\cos(\delta_i), \\ & v_{f,\varepsilon}^2 \sin(x_{3,i}), v_{f,\gamma}^2 \cos(x_{3,i})]^T \end{aligned} \tag{3d}$$

The corresponding coordinate transformation and feedback control law are

$$\begin{cases} \vartheta_{1,i}^X = x_{1,i} \\ \vartheta_{2,i}^X = x_{4,i}\cos(x_{3,i}) - x_{5,i}\sin(x_{3,i}) \\ \vartheta_{1,i}^Y = x_{2,i} \\ \vartheta_{2,i}^Y = x_{5,i}\cos(x_{3,i}) + x_{4,i}\sin(x_{3,i}) \\ \vartheta_{1,i}^\theta = x_{3,i} \\ \vartheta_{2,i}^\theta = x_{6,i} \end{cases} \tag{4}$$

$$\begin{aligned} u &= \tilde{A}(x_i)\nu_i \\ &= \begin{bmatrix} M_i\cos x_{3,i} & M_i\sin x_{3,i} & 0 \\ -M_i\sin x_{3,i} & M_i\cos x_{3,i} & 0 \\ 0 & 0 & I_i^z \end{bmatrix} \begin{bmatrix} \nu_i^X \\ \nu_i^Y \\ \nu_i^\theta \end{bmatrix} \end{aligned} \tag{5}$$

Different from the vehicle platoon driving on the straight lane, for the vehicle platoon on the curved road, to maintain the compactness, the arc-length-based spacing error rather than the longitudinal spacing error should be used to construct the vehicle platoon system. Therefore, the vehicle dynamics along the curved road should be obtained.

Suppose that the curved road is represented by the p-order polynomial, which is written as

$$Y(X) = a_0 X^p + a_1 X^{p-1} + \cdots + a_{p+1} \tag{6}$$

where X and Y are the horizontal and vertical coordinates; p is the positive integer; and a_k, $k = 0, \cdots, p+1$ are the coefficients. Herein, we use (6) to express that the curved road is due to arbitrary regular plane geometry can be approximated by a polynomial in theory.

According to (6), the distance along the curved road (the arc-length) driven by the ith following vehicle can be expressed as

$$S_i = \int_0^{X_i} \sqrt{1 + \beta(\tau)^2} d\tau \tag{7}$$

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where $\beta(\tau) = pa_0\tau^{p-1} + (p-1)a_1\tau^{p-2} + \cdots + a_p$. Differentiating S_i with respect to time and considering (2a), the vehicle dynamics model along the curved road can be obtained

$$\begin{cases} \dot{S}_i = \sqrt{1+\beta(\vartheta_{1,i}^X)^2}\vartheta_{2,i}^X \\ \dot{\vartheta}_{1,i}^X = \vartheta_{2,i}^X \\ \dot{\vartheta}_{2,i}^X = \nu_i^X + \mathcal{G}_i^X w_i^X \end{cases} \tag{8}$$

Obviously, (8) is also nonlinear. To linearize this equation, if the transformation of coordinates and the inputs are formulated by (9) and (10), then (8) can be linearized and re-expressed by (11).

$$\begin{cases} \xi_{1,i} = S_i \\ \xi_{2,i} = \sqrt{1+\beta(\vartheta_{1,i}^X)^2}\vartheta_{2,i}^X(t) \\ \xi_{3,i} = \vartheta_{1,i}^X \end{cases} \tag{9}$$

$$\nu_i^x = \frac{1}{\lambda_i}(-\kappa_i + \sigma_i) \tag{10}$$

where

$$\lambda_i = \sqrt{1+\beta(\xi_{3,i})^2}$$

$$\kappa_i = \frac{1}{\lambda_i^3}\xi_{2,i}^2\left(p(p-1)a_0\xi_{3,i}^{p-2} + \cdots + a_{p-1}\right)\beta\left(\xi_{3,i}\right)$$

$$\begin{cases} \dot{\xi}_{1,i} = \xi_{2,i} \\ \dot{\xi}_{2,i} = \sigma_i + \mathcal{G}_i^X \tilde{w}_i^X \\ \dot{\xi}_{3,i} = \xi_{2,i}/\lambda_i \end{cases} \tag{11}$$

where $\tilde{w}_i^X = \lambda_i w_i^X$.

B. Longitudinal Vehicle Platoon Modeling

For the platoon driven on the curved road, since the road is crooked, we adopt the arc-length rather than the linear segment to describe the intervehicle spacing errors (ISE). As shown in the coordinates in Fig. 1, the ISE and its derivative between the following vehicles i and $i-1$ is denoted by $\bar{S}_i$ and $\dot{\bar{S}}_i$, which is written as

$$\begin{cases} \bar{S}_i = S_{i-1} - S_i - S_d \\ \dot{\bar{S}}_i = \dot{S}_{i-1} - \dot{S}_i \end{cases} \tag{12}$$

where S_{i-1} and S_i are the arc-lengths that the vehicles $i-1$ and i traveled; and S_d is the ideal intervehicle spacing (IIS). To acquire the compact form of the platoon system, we define the new tracking error η_i for follower i as follows:

$$\eta_i = S_0 - S_i - \bar{S}_{d,i} = \sum_{j=1}^{i}\bar{S}_j \tag{13}$$

where $\bar{S}_{d,i} = iS_d$ is the IIS between the leader and the ith follower. Differentiating (13) with respect to time and considering (9) and (11), the following equation is obtained

$$\begin{cases} \dot{\eta}_i = \xi_{2,0} - \xi_{2,i} \\ \ddot{\eta}_i = \mathrm{Acc}_0 - \sigma_i - \mathcal{G}_i^X \tilde{w}_i^X \end{cases} \tag{14}$$

where Acc_0 means the acceleration of the leading vehicles. Considering that the leading vehicle can be manned in actuality, Acc_0 is nonzero and unknown for the following vehicles in this paper. Define $\Xi_i = [\,\eta_i\ \dot{\eta}_i\,]^T \in \mathbb{R}^{2\times 1}$, then (14) can be rewritten as

$$\dot{\Xi}_i(t) = A\Xi_i(t) + B\sigma_i + \hat{\mathcal{G}}_i\hat{w}_i \tag{15}$$

where $A = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$; $B = \left[\begin{smallmatrix} 0 \\ -1 \end{smallmatrix}\right]$;

$$\hat{\mathcal{G}}_i = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ -fg & -g & -C_\varepsilon/M_i & C_\gamma/M_i & 1 \end{bmatrix}; \hat{w}_i = \begin{bmatrix} \tilde{w}_i^X \\ \mathrm{Acc}_0 \end{bmatrix}.$$

In this paper, since the PLF topology is adopted, the control laws of the ith vehicle in (15) are expressed as

$$\begin{aligned} \sigma_i(t) = {} & k_{1,i}(\Xi_{1,i} - \Xi_{1,i-1}) + k_{2,i}(\Xi_{2,i} - \Xi_{2,i-1}) \\ & + k_{3,i}\Xi_{1,i} + k_{4,i}\Xi_{2,i} \end{aligned} \tag{16}$$

where $k_{j,i}$, $j = 1,2,3,4$ are the control gains to be determined.

Note that, in (16), the spacing errors $(\Xi_{1,i} - \Xi_{1,i-1})$ and the velocity error $(\Xi_{2,i} - \Xi_{2,i-1})$ come from the ISE between vehicles i and $i-1$, while the information $\Xi_{1,i}$ and $\Xi_{2,i}$ originate from the ISE between the leading vehicle and the ith vehicle. However, due to the discrete signals in the sensor and wireless communication and the data delay and dropouts in wireless communication, the platoon could be unstable [19].

To restrain those negative factors, we introduce two equivalent delays to quantify the discretization, delays, and dropouts of the data in the information transmitting procedure. Denote by ρ_s and ρ_w the minimal sample frequencies in the sensor and wireless communication. According to [22], the data delays generated by the discrete data in the sensor and wireless communication ς_t^s and ς_t^w satisfy

$$0 \le \varsigma_t^s \le 1/\rho_s,\ 0 \le \varsigma_t^w \le 1/\rho_w \tag{17}$$

Furthermore, in terms of [23], the data dropouts in the wireless communication can be approximated as the delay if the delay in communication is smaller than the sampling periods, and fulfills

$$\tau_d \le n_d T_d \tag{18}$$

where τ_d is the equivalent delay caused by the data dropouts; and n_d and T_d are, respectively, the maximum number of packet dropouts and the sampling periods. Let τ_t be the original delay; then, the total delay in communication can be expressed as

$$0 \le \varsigma_d \le \tau_t + n_d T_d \tag{19}$$

where ς_d means the equivalent delay in wireless communication.

Substituting (17) and (19) into (16), the control laws of the ith follower can be rewritten as

$$\begin{aligned} \sigma_i(t) = {} & k_{1,i}\left(\Xi_{1,i}(t-\varepsilon) - \Xi_{1,i-1}(t-\varepsilon)\right) \\ & + k_{2,i}\left(\Xi_{2,i}(t-\varepsilon) - \Xi_{2,i-1}(t-\varepsilon)\right) \\ & + k_{3,i}\Xi_{1,i}(t-\lambda) + k_{4,i}\Xi_{2,i}(t-\lambda) \end{aligned} \tag{20}$$

where $\varepsilon = t - \varsigma_t^s$ and $\lambda = \varsigma_d + t - \varsigma_t^w$.

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Combining (15) and (20), the model of the heterogeneous vehicle platoon can be represented by

$$\begin{cases} \dot{\Xi}(t) = \bar{A}\Xi(t) + \bar{B}K_1\mathcal{H}_1\Xi(t-\varepsilon) \\ \qquad\quad +\bar{B}K_2\mathcal{H}_2\Xi(t-\lambda) + \hat{\mathcal{G}}\hat{w} \\ Z(t) = C\Xi(t) \\ \Xi(t) = \varphi(t), \quad t \in [-h, 0] \end{cases} \tag{21}$$

where $\Xi = \mathrm{Col}[\Xi_i]_{i=1}^n$ is the state vector; $\hat{w} = \mathrm{Col}[\hat{w}_i]_{i=1}^n$ is the disturbance vector; $Z(t)$ is the control output; $h = \max\{h_1,\ h_2\}$, where $h_1 = 1/\rho_s$ and $h_2 = \tau_t + n_d T_d + 1/\rho_w$; $\phi(t)$ is the initial condition which is continuously differentiable when $t \in [-h,\ 0]$; and $\bar{A}, \bar{B}, \mathcal{H}_1, \mathcal{H}_2, C$ and $\hat{\mathcal{G}}$ are the coefficient matrices which are formulated by (22); and K_1 and K_2 are the gain matrices that can be represented by (23).

$$\bar{A} = I_n \otimes A;\ \bar{B} = I_n \otimes B;\ \hat{\mathcal{G}} = \mathrm{Col}\left[\hat{\mathcal{G}}_i\right]_{i=1}^n; C = I_n \otimes I_2;$$

$$\mathcal{H}_1 = \begin{bmatrix} I_2 & 0 & \cdots & 0 \\ -I_2 & I_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & -I_2 & I_2 \end{bmatrix};$$

$$\mathcal{H}_2 = \begin{bmatrix} I_2 & 0 & \cdots & 0 \\ 0 & I_2 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & I_2 \end{bmatrix} \tag{22}$$

$$K_1 = \mathrm{diag}\,\{K_{i,1}\}_{i=1}^n$$
$$K_2 = \mathrm{diag}\,\{K_{i,2}\}_{i=1}^n \tag{23}$$

where the symbol $\otimes$ means the Kronecker product; I_k, $k = 2$ or n is the k-order identity matrix; $K_{i,1} = [\,k_{1,i}\ k_{2,i}\,]$ and $K_{i,2} = [\,k_{3,i}\ k_{4,i}\,]$.

C. Lateral Lane Keeping Model

In the modeling of the vehicle platoon in section II-B, since only the longitudinal subsystem of the vehicle in (2a) is used to establish the vehicle platoon model, the lane keeping ability of the following vehicles for preventing collisions with the cars in the adjacent lane might not be satisfied. To avoid lane departure, the lateral lane keeping model of the following vehicle is constructed.

Fig. 2 shows the relationship between the ith follower and its ideal position point when it deviates from the centerline of the curved road. In this figure, point P_{idea} denotes the ideal location of the ith following vehicle on the curved road, which can be confirmed by the lagging distance $\bar{S}_{d,i}$ from the leader along the curved road. The symbols $\hat{\vartheta}_i^Y$ and $\hat{\vartheta}_i^\theta$ are the ideal lateral position and tangential angle of the ith following vehicle on point P_{idea}, which are expressed as

$$\begin{cases} \hat{\vartheta}_i^Y = Y(\tilde{X}(S_0 - \bar{S}_{d,i})) \\ \hat{\vartheta}_i^\theta = \arctan\left(\frac{d\hat{\vartheta}_i^Y}{dX}\right) \end{cases} \tag{24}$$

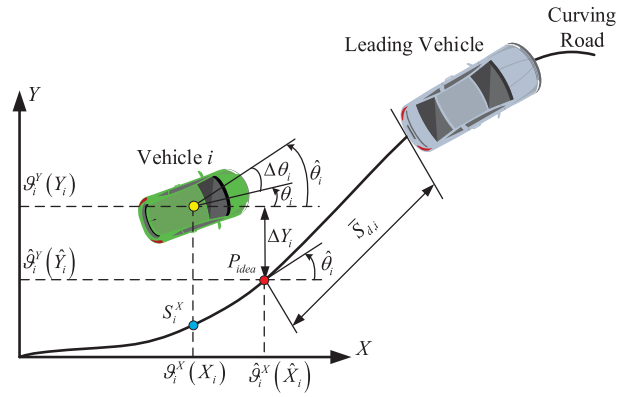

Fig. 2. The tracking errors between the ith following vehicle and its ideal position on the curving road.

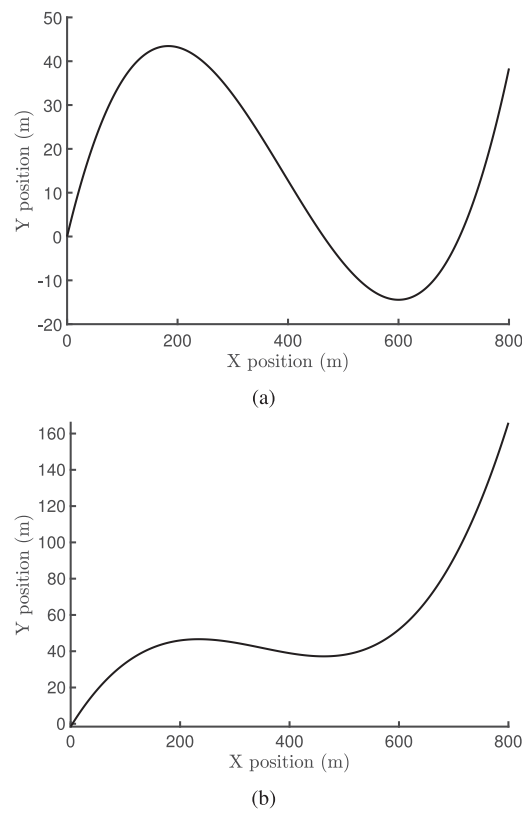

Fig. 3. The shapes of curving roads. (a) Curve I: cubic polynomial. (b) Curve II: 4th degree polynomial.

where $\tilde{X}(\cdot)$ is the inverse function of (7) and $\arctan(\cdot)$ means the arctangent.

According to the geometrical relationship between the ith follower and point P_{idea} in Fig. 3, and considering the subsystems in (2b) and (2c), the tracking errors model of the ith following vehicle can be written as

$$\begin{cases} \Delta\dot{Y}_i = \hat{\vartheta}_{2,i}^Y - \vartheta_{2,i}^Y \\ \Delta\ddot{Y}_i = \hat{\nu}_i^Y - \nu_i^Y - \mathcal{G}_i^Y w_i^Y \end{cases} \tag{25a}$$

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$$\begin{cases} \Delta\dot{\theta}_i = \hat{\vartheta}^{\theta}_{2,i} - \vartheta^{\theta}_{2,i} \\ \Delta\ddot{\theta}_i = \hat{\nu}^{\theta}_i - \nu^{\theta}_i \end{cases} \tag{25b}$$

where $\Delta Y = \hat{\vartheta}^Y_{1,0} - \vartheta^Y_{1,i}$ is the lateral distance errors between the ith following vehicle and the point P_{idea} in the inertial frame; $\Delta\theta = \hat{\vartheta}^{\theta}_{1,0} - \vartheta^{\theta}_{1,i}$ is angle errors between the heading angle of the ith following vehicle and the tangential angle of point P_{idea}; $\hat{\nu}^Y_i$ and $\hat{\nu}^{\theta}_i$ are the lateral and angle accelerations of the ith following vehicle on the point P_{idea}, which are unknown but can be deemed as the disturbances; and $\hat{\vartheta}^Y_{2,i}$ and $\hat{\vartheta}^{\theta}_{2,i}$ are the lateral and tangential-angle speeds of the ith following vehicle on the point P_{idea}, which can be formulated by

$$\begin{cases} \hat{\vartheta}^Y_{2,i} = v_0 \sin(\hat{\vartheta}^{\theta}_i) \\ \hat{\vartheta}^{\theta}_{2,i} = \frac{v_0}{\rho_i} \end{cases} \tag{26}$$

where v_0 is the velocity of the leading vehicle; and ρ_i is the curve radius of the curving road at the point P_{idea}.

Denote $K^Y_i = [\,k^Y_1\ k^Y_2\,]$ and $K^{\theta}_i = [\,k^{\theta}_1\ k^{\theta}_2\,]$ as the control gains of the error systems in (25); then, the inputs ν^Y_i and ν^{θ}_i can be indicated as

$$\begin{cases} \nu^Y_i = k^Y_1 \Delta Y + k^Y_2 \Delta\dot{Y} \\ \nu^{\theta}_i = k^{\theta}_1 \Delta\theta + k^{\theta}_2 \Delta\dot{\theta} \end{cases} \tag{27}$$

Let $\Gamma^Y_i = [\,\Delta Y_i\ \Delta\dot{Y}_i\,]^T \in \mathbb{R}^{2\times 1}$ and $\Gamma^{\theta}_i = [\,\Delta\theta_i\ \Delta\dot{\theta}_i\,]^T \in \mathbb{R}^{2\times 1}$ be the states of the error systems. Combined with (27), the lateral tracking error model in (25) can be re-expressed as

$$\begin{cases} \dot{\Gamma}^Y(t) = \bar{A}\Gamma^Y(t) + \bar{B}K^Y\Gamma^Y(t) + \hat{\mathcal{G}}^Y\hat{W}^Y \\ \Upsilon^Y(t) = C\Gamma^Y(t) \end{cases} \tag{28a}$$

$$\begin{cases} \dot{\Gamma}^{\theta}(t) = \bar{A}\Gamma^{\theta}(t) + \bar{B}K^{\theta}\Gamma^{\theta}(t) + \hat{\mathcal{G}}^{\theta}\hat{W}^{\theta} \\ \Upsilon^{\theta}(t) = C\Gamma^{\theta}(t) \end{cases} \tag{28b}$$

where $\Gamma^{\dagger} = \mathrm{Col}[\Gamma^{\dagger}_i]^n_{i=1}$ and symbol $\dagger$ means Y and θ; $K^{\dagger} = \mathrm{diag}\{[k^{\dagger}_1, k^{\dagger}_2]\}^n_{i=1}$; $\hat{\mathcal{G}}^{\dagger} = \mathrm{Col}[\hat{\mathcal{G}}^{\dagger}_i]^n_{i=1}$,

$$\hat{\mathcal{G}}^Y_i = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ -g & -fg & -C_{\varepsilon}/M_i & -C_{\gamma}/M_i & 1 \end{bmatrix}$$

and $\hat{\mathcal{G}}^{\theta}_i = \left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$; $\hat{W}^{\dagger} = \mathrm{Col}\,[\hat{w}^{\dagger}_i]^n_{i=1}$, $\hat{w}^Y_i = \left[\begin{smallmatrix} w^Y_i \\ \hat{\nu}^Y_i \end{smallmatrix}\right]$ and $\hat{w}^{\theta}_i = \hat{\nu}^{\theta}_i$.

III. Robust Control of the Platoon

A. Control Objectives

Insomuch as the disturbances exist, the performance of the interference rejection must be satisfied. Therefore, the following performances are given

$$J(\hat{w}) = \int_0^{\infty} \Big((Z(t))^T Z(t) - \gamma_1^2 (\hat{w}(t))^T \hat{w}(t) \Big) dt \tag{29a}$$

$$J^{\dagger}(\hat{W}^{\dagger}) = \int_0^{\infty} \left(\left(\Upsilon^{\dagger}(t)\right)^T \Upsilon^{\dagger}(t) - \left(\gamma^{\dagger}_2\right)^2 \left(\hat{W}^{\dagger}(t)\right)^T \hat{W}^{\dagger}(t) \right) dt \tag{29b}$$

where γ_1 and $\gamma^{\dagger}_2$ are constants.

To address the information delays and external disturbances in the vehicle platoon, robust control approaches are adopted, and the control objectives are

1) The closed-loop systems in (21) and (28) are asymptotically stable when the disturbances are equal to zero, i.e., the following conditions are satisfied,

$$\lim_{t\to\infty} \|\Xi_i\| = 0,\ \lim_{t\to\infty} \left\|\dot{\Xi}_i\right\| = 0 \tag{30a}$$

$$\lim_{t\to\infty} \left\|\Gamma^{\dagger}_i\right\| = 0,\ \lim_{t\to\infty} \left\|\dot{\Gamma}^{\dagger}_i\right\| = 0 \tag{30b}$$

2) $J(\hat{w})$ and $J^{\dagger}(\hat{W}^{\dagger})$ in (29) are less than zero for all non-zero $\hat{w}(t)$ and $\hat{W}^{\dagger}(t)$ that belong to the $\mathcal{L}_2$-norm space and the given positive constants γ_1 and $\gamma^{\dagger}_2$ under the zero initial condition.
3) The transient deviations are not amplified when they are propagated downstream along the platoon for any maneuver of the leading vehicle.

The first and second conditions in the above control objectives are to ensure the intervehicle stability of the vehicle platoon and impel that the tracking errors in (25) converge to zero, and the last condition is the requirement of assuring the string stability of the vehicle platoon.

B. Platoon Stable Controller Design

Since the communication delays and disturbances exist in the vehicle platoon system in (21), the H_∞ method based on the Lyapunov-Krasovskii functional is adopted to maintain the intervehicle stability of the vehicle platoon. Before giving the main results on the controller design, we first give a Lemma, which will play a vital role in the main results.

Lemma 1: For any matrices $\mathcal{X} \geq 0$, $U = U^T > 0$, and $Z = Z^T > 0$, and for any appropriately dimensioned matrices $\mathcal{N}$, the sufficient condition of nonlinear matrix inequality (31) holding is the linear matrix inequality (32) being satisfied.

$$\begin{bmatrix} \mathcal{X} & \mathcal{N} \\ * & UZU \end{bmatrix} \geq 0 \tag{31}$$

$$\begin{bmatrix} \mathcal{X} & \mathcal{N} \\ * & 2U - \mathcal{Z} \end{bmatrix} \geq 0 \tag{32}$$

where $*$ in the matrix denotes the symmetric elements, and $\mathcal{Z}$ is the inverse matrix of Z.

Proof: Since $U = U^T > 0$ and $Z = Z^T > 0$, the following inequality can be fulfilled:

$$(U - \mathcal{Z})Z(U - \mathcal{Z}) \geq 0 \tag{33}$$

According to (33), we can obtain

$$UZU \geq 2U - \mathcal{Z} \tag{34}$$

Note that $\mathcal{X} \geq 0$ and in the light of LMI (34), we can know that if LMI (32) holds, the nonlinear matrix inequality (31) is obtained. ■

Based on Lemma 1, the sufficient conditions that can ensure the intervehicle stability of the vehicle platoon is given by the following theorem.

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Theorem 1: Consider the closed-loop heterogeneous vehicle platoon system on the curved road in (21). For given time delays $h_1 \geq 0$ and $h_2 \geq 0$, and the scalar $\gamma_1 \geq 0$, the system is asymptotically stable and satisfies $J(\hat{w}) < 0$ for all nonzero $\int_0^\infty \|\hat{w}(t)\|^2 dt < \infty$ under the condition $\Xi(t) = 0, \forall t \in [-h,\ 0]$, if there exist matrices $\mathcal{P} > 0, \mathcal{K}_m, \mathcal{Q}_m > 0, m = 1, 2$, and $\varpi_{tt} > 0$, $tt = 1, 2, 3$, and any appropriately dimensioned matrices $\tilde{\mathcal{D}}, \tilde{\mathcal{U}}, \tilde{\mathcal{R}}, \tilde{\mathcal{X}}, \tilde{\mathcal{F}}$, and $\tilde{\mathcal{O}}$, such that the following LMIs hold:

$$\begin{bmatrix} \tilde{\Phi}_2 & \sqrt{h_1}(\bar{\mathcal{C}})^T & \sqrt{h_2}(\bar{\mathcal{C}})^T & \sqrt{h_3}(\bar{\mathcal{C}})^T & \Sigma^T \mathcal{I}_1^T C^T \\ * & -\varpi_1 & 0 & 0 & 0 \\ * & * & -\varpi_2 & 0 & 0 \\ * & * & * & -\varpi_3 & 0 \\ * & * & * & * & -I \end{bmatrix} \leq 0 \tag{35}$$

$$\begin{bmatrix} \tilde{\mathcal{X}} & \tilde{\mathcal{D}} \\ * & 2\mathcal{P} - \varpi_1 \end{bmatrix} \geq 0 \tag{36}$$

$$\begin{bmatrix} \tilde{\mathcal{F}} & \tilde{\mathcal{U}} \\ * & 2\mathcal{P} - \varpi_2 \end{bmatrix} \geq 0 \tag{37}$$

$$\begin{bmatrix} \tilde{\mathcal{O}} & \tilde{\mathcal{R}} \\ * & 2\mathcal{P} - \varpi_3 \end{bmatrix} \geq 0 \tag{38}$$

where $h_3 = |h_1 - h_2|$, $\tilde{\Phi}_2 = \tilde{\Phi}_1 + \{\tilde{\Phi}_3\}_{sym} + h_1\tilde{\mathcal{X}} + h_2\tilde{\mathcal{F}} + h_3\tilde{\mathcal{O}}$, $\{\cdot\}_{sym}$ is the sum of a matrix and its transposed matrix,

$$\tilde{\Phi}_1 = \begin{bmatrix} \bar{A}\mathcal{P} + \mathcal{P}\bar{A}^T + \sum_{j=1}^2 \mathcal{Q}_j & \bar{B}\mathcal{K}_1 & \bar{B}\mathcal{K}_2 & 0 & 0 & \hat{\mathcal{G}} \\ * & 0 & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 & 0 \\ * & * & * & -\mathcal{Q}_1 & 0 & \\ * & * & * & * & -\mathcal{Q}_2 & 0 \\ * & * & * & * & * & -\gamma_1^2 I \end{bmatrix},$$

$$\tilde{\Phi}_3 = \begin{bmatrix} \tilde{\mathcal{D}} + \tilde{\mathcal{U}} & 0 & 0 & -\tilde{\mathcal{D}} - \tilde{\mathcal{R}} & \tilde{\mathcal{R}} - \tilde{\mathcal{U}} & 0 \end{bmatrix},$$

$$\Sigma = \mathrm{diag}\{\mathcal{P},\ \mathcal{P},\ \mathcal{P},\ \mathcal{P}, \mathcal{P},\ I\},$$

$$\bar{\mathcal{C}} = \begin{bmatrix} \bar{A}\mathcal{P} & \bar{B}\mathcal{K}_1 & \bar{B}\mathcal{K}_2 & 0 & 0 & \hat{\mathcal{G}} \end{bmatrix},$$

$$\mathcal{I}_1 = \begin{bmatrix} I_{2n} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Furthermore, the feedback gain matrices K_1 and K_2 can be obtained by

$$\begin{cases} K_1 = \mathcal{K}_1(\mathcal{P})^{-1}\mathcal{H}_1^{-1} \\ K_2 = \mathcal{K}_2(\mathcal{P})^{-1}\mathcal{H}_2^{-1} \end{cases} \tag{39}$$

Proof: The proof is given in Appendix A. ■

C. Robust Tracking Controller Design

To eliminate the lateral errors of the following vehicle under the disturbance and satisfy the control performance in (29b), the robust tracking controller based on H_∞ is established in this section.

Theorem 2: Consider the tracking error systems described in (28). For given scalars $\gamma_2^\dagger$, the system is stable and satisfies $J^\dagger(\hat{W}^\dagger) < 0$ for all nonzero $\int_0^\infty \|\hat{W}^\dagger(t)\|^2 dt < \infty$, if there exist matrices $\hat{P}^\dagger > 0$, $\hat{K}^\dagger$, such that the following LMI hold:

$$\begin{bmatrix} \hat{P}^\dagger \bar{A}^T + \bar{A}\hat{P}^\dagger + (\bar{B}\hat{K}^\dagger)^T + \bar{B}\hat{K}^\dagger & \hat{\mathcal{G}}^\dagger & \hat{P}^\dagger C^T \\ * & -\gamma_2^\dagger I & 0 \\ * & * & -\gamma_2^{\dagger I} \end{bmatrix} < 0 \tag{40}$$

If LMI (40) is feasible, then the control gains of the systems in (28) can be written as

$$K^\dagger = \hat{K}^\dagger \left(\hat{P}^\dagger\right)^{-1} \tag{41}$$

Proof: According to the Bounded Real Lemma in [24], for the tracking error model in (28), the following matrix inequality is obtained:

$$\begin{bmatrix} (\bar{A} + \bar{B}K^\dagger)^T P^\dagger + P^\dagger(\bar{A} + \bar{B}K^\dagger) & P^\dagger\hat{\mathcal{G}}^\dagger & C^T \\ * & -\gamma_2^\dagger I & 0 \\ * & * & -\gamma_2^\dagger I \end{bmatrix} < 0 \tag{42}$$

where $P^\dagger > 0$ is a symmetric matrix. Pre-multiply and post-multiply (42) by $\mathrm{diag}\{(P^\dagger)^{-1},\ I,\ I\}$, and let $\hat{P}^\dagger = (P^\dagger)^{-1}$, $\hat{K}^\dagger = K^\dagger\hat{P}^\dagger$; then, LIM (40) is obtained. ■

IV. String Stability Analysis

To guarantee the string stability of the vehicle platoon under the disturbances and communication delay, the definition of the $\mathcal{L}_2$ string stability in [25] is used to propose the stability criterion.

Definition 1 ([25]): Consider the following time-delay nonautonomous platoon system

$$\begin{aligned} \dot{\chi}_i(t) &= f_e\left(\chi_i(t-\tau_1), \chi_{i-1}(t-\tau_2), w_i(t)\right) \\ y_i(t) &= f_h\left(\chi_i(t)\right), \\ \chi_i(t) &= \varphi_i(t), \quad i \in \mathcal{I}_N,\ t \in [-\tau, 0], \end{aligned} \tag{43}$$

where $\mathcal{I}_N = \{1, \cdots, N\}$ means the N following vehicles; $\chi_i \in \mathbb{R}^{n_\chi}$ is the state vector; $w_i \in \mathbb{R}^{n_w}$ is the external input; $y_i \in \mathbb{R}^m$ is the output; τ_1 and τ_2 are the time delays; and $\tau = \max\{\tau_1, \tau_2\}$. Moreover, $f_e : \mathbb{R}^n \times \mathbb{R}^n \mapsto \mathbb{R}^n$; $f_h : \mathbb{R}^{n_\chi} \mapsto \mathbb{R}^m$; and $\varphi_i : \mathbb{R}^1 \mapsto \mathbb{R}^{n_\chi}$ is continuously differentiable for $t \in [-\tau,\ 0]$. Let $\bar{\chi}_i$ be the equilibrium solution of (43) for $w_i = 0$. The system (43) is $\mathcal{L}_2$ string stable if there exist class $\mathcal{K}$ functions $\bar{\alpha}$ and $\bar{\beta}$ such that, for any initial state $\varphi_i(t)$, $t \in [-\tau,\ 0]$, and any $w_i(t) \in \mathcal{L}_2^{n_w}$, we have

$$\|y_i(t) - f_h(\bar{\chi}_i)\|_{\mathcal{L}_2} \leq \bar{\alpha}\left(\|w_i(t)\|_{\mathcal{L}_2}\right) + \bar{\beta}\left(\|\varphi_i(t) - \bar{\chi}_i\|\right). \tag{44}$$

where $\|\cdot\|_{\mathcal{L}_2} = \sqrt{\int_0^\infty \|\cdot\|^2 dt} < \infty$ is the $\mathcal{L}_2$ norm. If, in addition, with $\bar{\chi}_i$ it also holds that

$$\|y_i(t) - f_h(\bar{\chi}_i)\|_{\mathcal{L}_2} \leq \|y_{i-1}(t) - f_h(\bar{\chi}_i)\|_{\mathcal{L}_2}, \tag{45}$$

and system (43) is strictly $\mathcal{L}_2$ string stable.

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Based on Definition 1 and Theorem 1, the criterion that ensures the string stability of the vehicle platoon system on the curved road is given.

Theorem 3: Consider the platoon dynamics described in (21). The vehicle platoon system is $\mathcal{L}_2$ string stable under the disturbances and information delays, by using the feedback gains K_1 and K_2 acquired from (39), if the LMIs (35)–(38) are feasible. Moreover, if the LMIs (35)–(38) are also feasible for all $0 < \gamma_1 < 1$, then the vehicle platoon in (21) are strictly $\mathcal{L}_2$ string stable.

Proof: According to Theorem 1, the feasibility of the LMIs (35)–(38) implies that $J(\hat{w}) < 0$, i.e., $\|Z(t)\|_{\mathcal{L}_2} < \gamma_1 \|\hat{w}(t)\|_{\mathcal{L}_2}$. In addition, it is obvious that $\Xi_i(0) = 0$ is one equilibrium solution of (21), and thus, the condition in (44) is satisfied.

Moreover, let $\mathcal{T}_{Z_i\hat{w}_i}$ be the transfer function from disturbance $\hat{w}_i$ to output Z_i, and let $\mathcal{T}_{Z_i\Xi_{i-1}}$ be the transfer function from Ξ_{i-1} to Z_i. According to (15) and (20), we can gain

$$\tilde{Z}_i(s) = \mathcal{T}_{Z_i\hat{w}_i}\tilde{w}_i(s) + \mathcal{T}_{Z_i\Xi_{i-1}}\tilde{\Xi}_{i-1}(s) \tag{46}$$

where $\tilde{Z}_i$, $\tilde{w}_i$ and $\tilde{\Xi}_{i-1}$ are the corresponding Laplace transform of Z_i, w_i and Ξ_{i-1}.

From (46), it can be seen that $\mathcal{T}_{Z_i\hat{w}_i}$ is the complementary sensitivity of $\mathcal{T}_{Z_i\Xi_{i-1}}$. Because $\|Z(t)\|_{\mathcal{L}_2} < \gamma_1\|\hat{w}(t)\|_{\mathcal{L}_2}$, we can obtain

$$\|\mathcal{T}_{Z_i\hat{w}_i}(s)\|_\infty = \sup_{\tilde{w}_i \neq 0} \frac{\left\|\tilde{Z}_i\right\|_{\mathcal{L}_2}}{\|\tilde{w}_i\|_{\mathcal{L}_2}} < \gamma_1. \tag{47}$$

According to (47) and the Nyquist plot of $\mathcal{T}_{Z_i\hat{w}_i}(s)$, one can know that $\mathcal{T}_{Z_i\hat{w}_i}(jw)$ is located in a circle with the origin as the center and the radius γ_1. Therefore, if $0 < \gamma_1 < 1$, $\mathcal{T}_{Z_i\Xi_{i-1}}(s)$ satisfies

$$\|\mathcal{T}_{Z_i\Xi_{i-1}}(s)\|_\infty = \|I - \mathcal{T}_{Z_i\hat{w}_i}(s)\|_\infty \leq 1. \tag{48}$$

In addition, because output matrix C in (21) is a unit matrix, the transfer function from Ξ_{i-1} to Ξ_i is also equal to $\mathcal{T}_{Z_i\Xi_{i-1}}(s)$. Consequently, $\|\tilde{\Xi}_i\|_{\mathcal{L}_2} \leq \|\tilde{\Xi}_{i-1}\|_{\mathcal{L}_2}$; in other words, the condition of strict $\mathcal{L}_2$ string stability in (45) is fulfilled. ■

V. Simulation and Analysis

To verify the effectiveness of the proposed robust controller, a simulation about a heterogeneous vehicle platoon that consists of one leader and five followers driving along curved roads is carried out. To test the adaptability of the proposed control method to the curved roads with different shapes, two curved roads, called Curve I and Curve II, which can be represented by the cubic and 4th-degree polynomials respectively, are adopted. The expressions of the two curved roads are given in (49), and the shapes of the roads are shown in Fig. 3, where the lengths of the roads are 821 m.

$$Y(X) = a_0(X + X_0)^3 + a_1(X + X_0)^2 + a_2(X + X_0) + a_3 \tag{49a}$$

$$Y(X) = b_0(X + X_0)^4 + b_1(X + X_0)^3 + b_2(X + X_0)^2 + b_3(X + X_0) + b_4 \tag{49b}$$

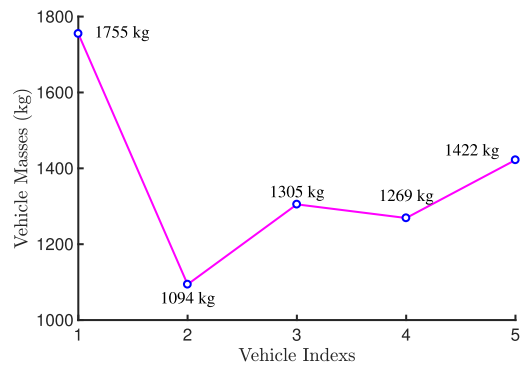

Fig. 4. The masses of following vehicles.

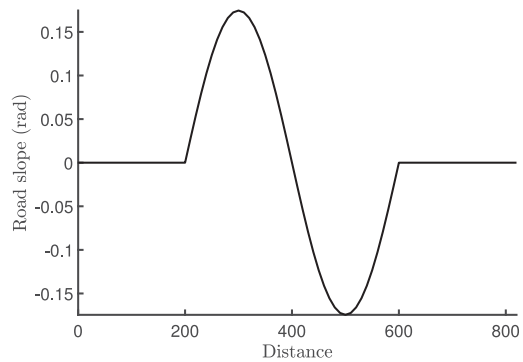

Fig. 5. The slope of curving road.

where $a_0 = 1.6 \times 10^{-6}$; $a_1 = 1 \times 10^{-3}$; $a_2 = -6 \times 10^{-5}$; $a_3 = -14.436$; $X_0 = -600$; $b_0 = 1.6 \times 10^{-10}$; $b_1 = 1.6 \times 10^{-6}$; $b_2 = 1.6 \times 10^{-4}$; $b_3 = 1 \times 10^{-2}$; and $b_4 = 40.59$.

Since the vehicle platoon is heterogeneous, the mass of each vehicle, which is given in Fig. 4, is different. In addition, the road's slope is treated as an external disturbance, as shown in Fig. 5. Herein, to simplify the simulation, the slopes of Curves I and II are assumed to be the same. Besides, since precise wind flow models are difficult to obtain, the longitudinal and lateral wind speeds are presented by white Gaussian noise in Fig. 6. The maximum packet dropout and the time-delay in the wireless communication are $n_d = 2$ and $\tau_t = 0.3$ s, and their variations conform to uniform distributions. The sample frequencies of radar and wireless communication are $\rho_s = 10$ Hz and $\rho_w = 20$ Hz, respectively.

During the simulation, the leading vehicle is hypothesized to drive along the road all of the time. To show the behavior of the vehicle platoon in the actual traffic environment, the Highway Fuel Economy Test (HWFET), which is the standard highway operation of a vehicle, is applied to generate the speed profiles of the leading vehicle running along Curve I and Curve II, as shown in Fig. 7. Moreover, the initial velocities of the following vehicles on Curve I and Curve II are 67 km/h and 79 km/h, and the first arc-lengths of the vehicles in the platoon are, respectively, 60 m, 50 m, 40 m, 30 m, 20 m, and 10 m.

To demonstrate the superiorities of the propounded robust control technique in handling communication delays and interference, the traditional string stability control in [26], [27] which is useful in the scenario of absence of interference and

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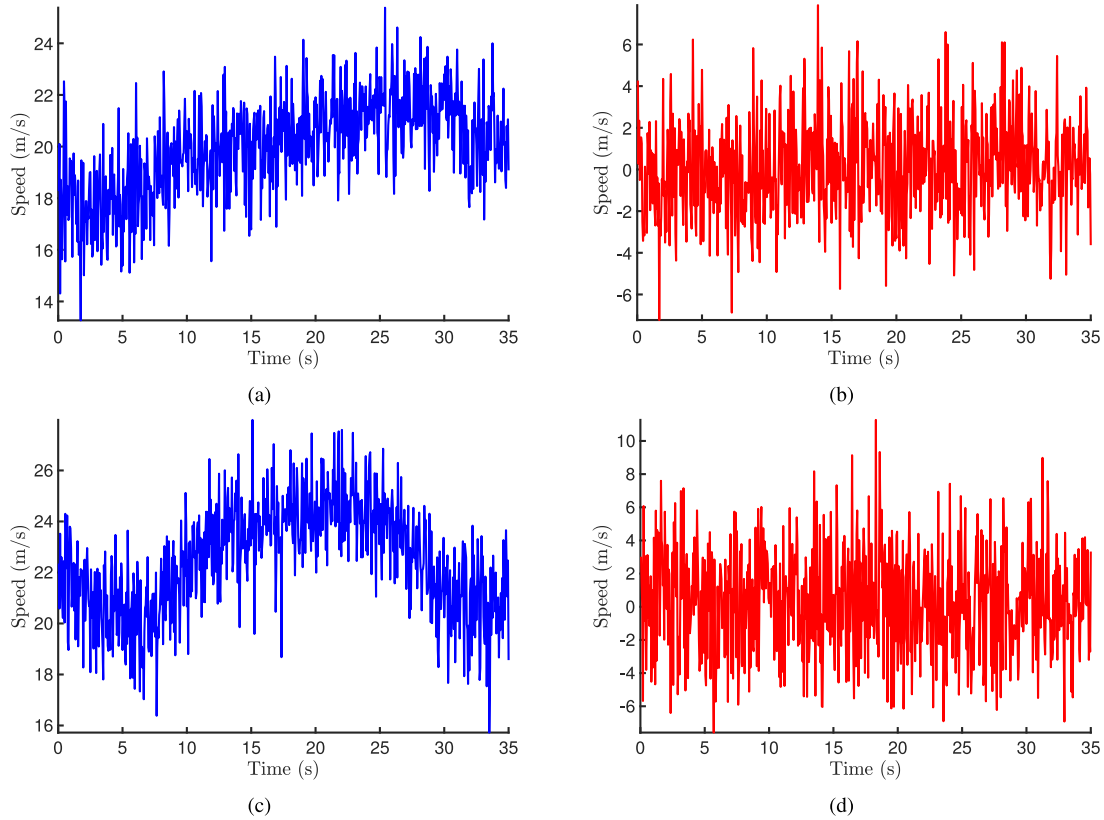

Fig. 6. The wind speeds during platoon running on curving roads. (a) Longitudinal wind speed when vehicles on Curves I. (b) Lateral wind speed when vehicles on Curves I. (c) Longitudinal wind speed when vehicles on Curves II. (d) Lateral wind speed when vehicles on Curves II.

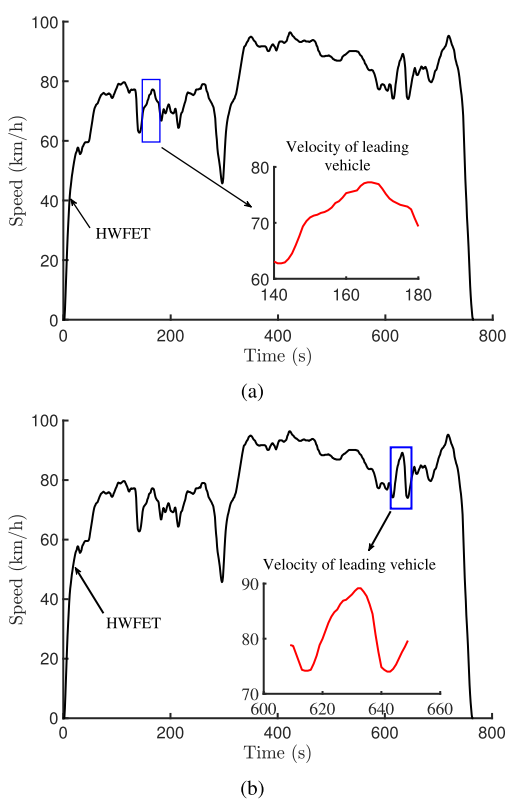

Fig. 7. The speed profile of the leading vehicle. (a) Curve I. (b) Curve II.

communication delay is employed to be the comparison and considered to be the regular control in this paper.

The results of the platoon simulation are shown in Figs. 8–11. Fig. 8 shows the spacing errors between adjacent following vehicles along the curved roads. By observing the spacing errors under robust control, which are shown in Figs. 8(a) and 8(c), it can be seen that, regardless of whether we use Curves I or II, the spacing errors between adjacent following vehicles in the robust control model approach zero at the end of the simulation. Moreover, the amplitudes of the spacing errors are decreased from Vehicle 1 to Vehicle 5, which indicates the damping of the transient deviations, in other words, the string stability of the platoon is satisfied. However, for the regular control in Figs. 8(b) and 8(d), the spacing errors between adjacent following vehicles for both Curves I and II are increased when the vehicle index enlarges, which means that the string stability is not fulfilled.

Fig. 9 shows the longitudinal velocities of the vehicles and the maximum speed error $E_{v,m}$ of a following vehicle with respect to the leading vehicle. From this figure, it can be seen that compared with the regular control, the following vehicles in the robust controller, for both Curves I and II, can track the speeds of the leading vehicle accurately, and the maximum of $E_{v,m}$ in the robust control for Curve I and Curve II are, respectively, 0.5 km/h and 1.2 km/h. By considering the magnitude of the leading vehicle speed, we can say that the proposed robust control method can realize a precision speed-tracking control. Moreover, we note that phenomenon of errors that amplify

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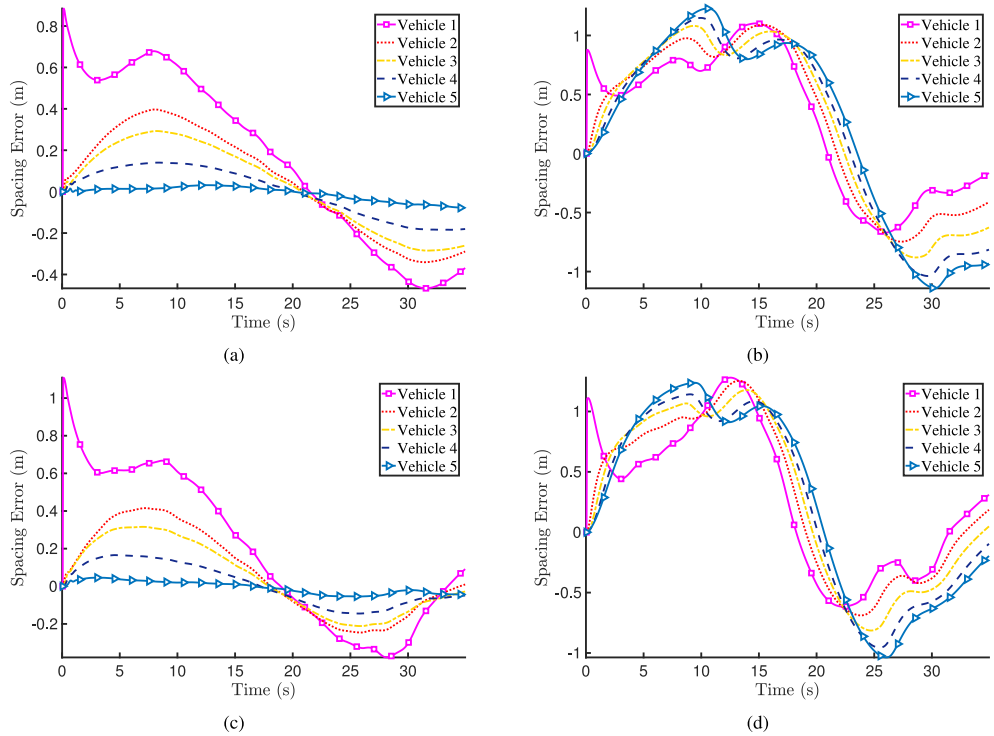

Fig. 8. The spacing errors of vehicles on curving roads. (a) Robust control when vehicles on Curve I. (b) Regular control when vehicles on Curve I. (c) Robust control when vehicles on Curve II. (d) Regular control when vehicles on Curve II.

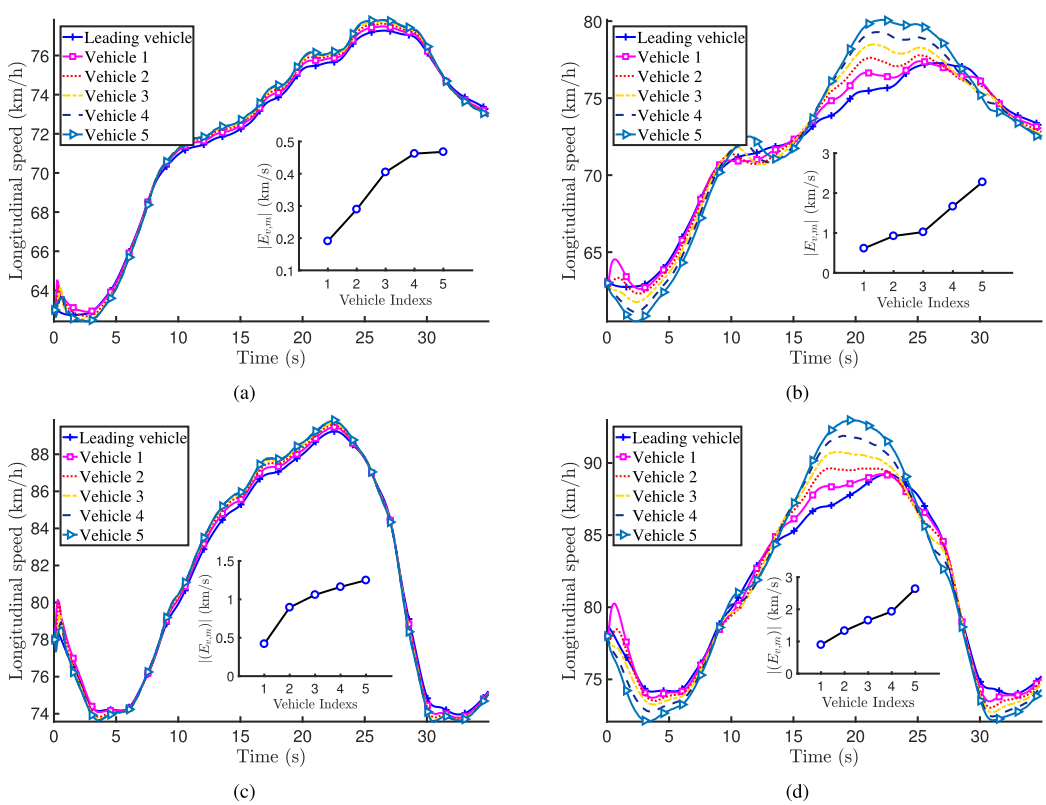

Fig. 9. The longitudinal velocities of following vehicles on curving roads. (a) Robust control when vehicles on Curve I. (b) Regular control when vehicles on Curve I. (c) Robust control when vehicles on Curve II. (d) Regular control when vehicles on Curve II.

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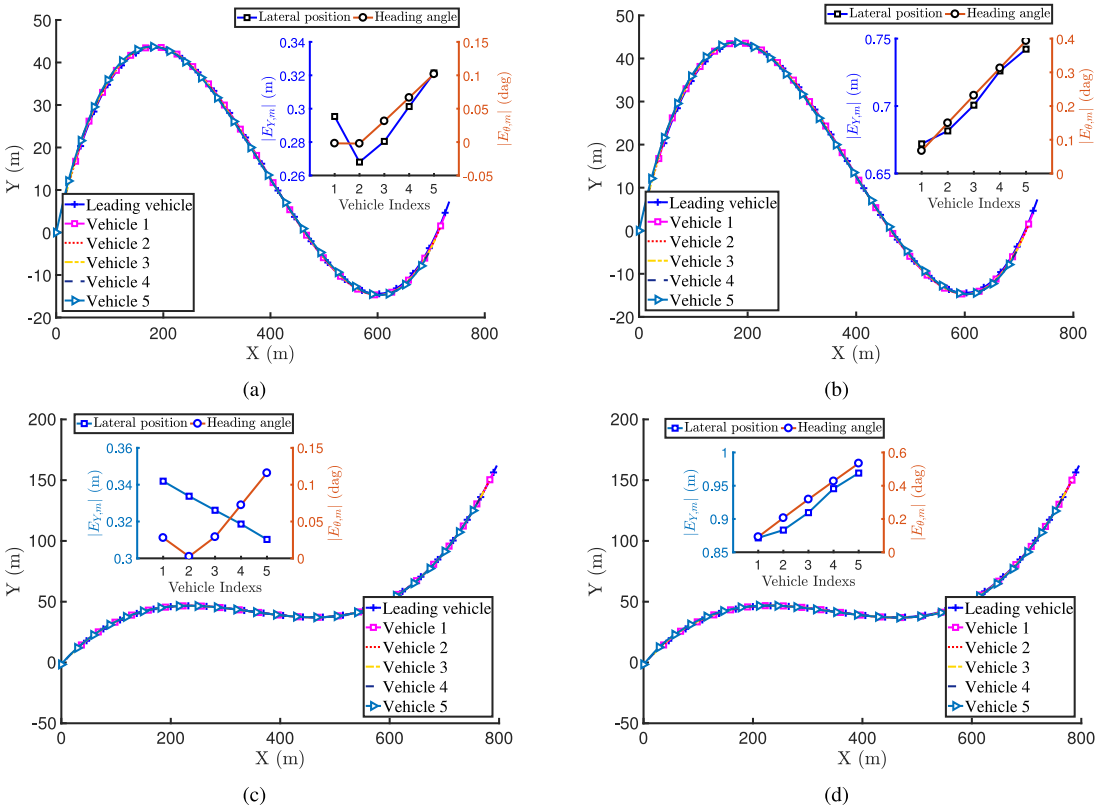

Fig. 10. The trajectories of the vehicles in the platoon. (a) Robust control during vehicles on Curve I. (b) Regular control during vehicles on Curve I. (c) Robust control during vehicles on Curve II. (d) Regular control during vehicles on Curve II.

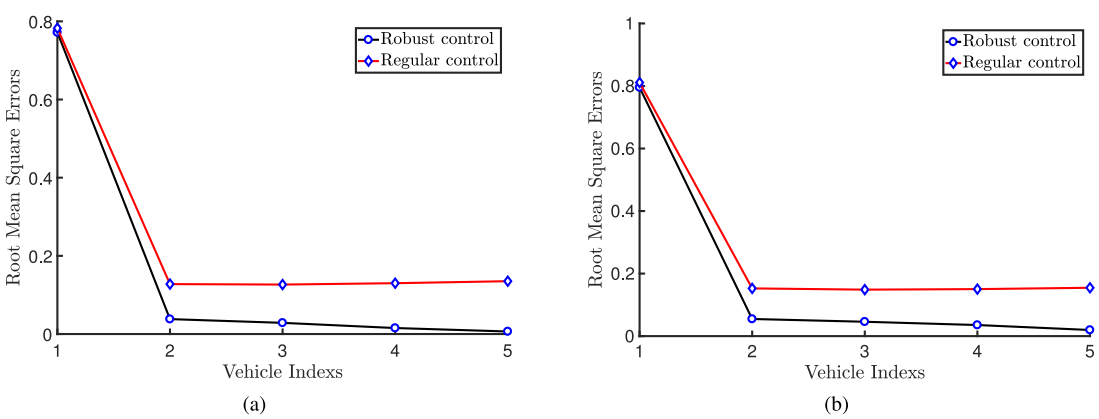

Fig. 11. The RMS speed errors of adjacent vehicles in the platoon. (a) Curve I. (b) Curve II.

with the increasing vehicle index in Fig. 9 is due to the error accumulation. This conclusion is verifiable from Fig. 11, where the root-mean-square (RMS) errors of the longitudinal velocity about the adjacent vehicles are provided. From the figure, it can be viewed that, for Curves I and II, the longitudinal velocities RMS errors of contiguous vehicles in both the regular and robust controls are degenerative as the vehicle index increases. In comparison with the regular control, the velocity RMS errors in the robust control, nevertheless, are minor.

The trajectories of the vehicles during the simulation are presented in Fig. 10, where the $E_{Y,m}$ and $E_{\theta,m}$ are the maximum lateral-position errors and heading-angle errors of the following vehicles with respect to the curved road respectively. By observing this figure, it can be seen that the $E_{Y,m}$ and $E_{\theta,m}$ of the following vehicles in the robust and regular control for Curves I are, respectively, 0.32 m and 0.1 deg, and 0.75 m and 0.4 deg. In addition, the $E_{Y,m}$ and $E_{\theta,m}$ of the following vehicles in the robust and regular controls for Curves II are, respectively, 0.34 m and 0.12 deg, and 1 m and 0.6 deg. Since the tracking errors in the robust control are smaller than in the regular control, accurate road tracking can be achieved in the proposed controller.

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VI. Conclusion

This paper proposes a novel string stability control technique and modeling for heterogeneous vehicle platoons on curved routes with up-down gradients while sustaining wireless communication delays and stochastic wind. Based on the technique of nonlinear system feedback linearization and having the arc length of an arbitrarily curved road represented by the polynomial, the vehicle platoon system is divided into a longitudinal vehicle platoon system along the curved road and a lateral road tracking system. To assure the string stability of the vehicle platoon system and to maintain the lane adherence of the vehicles, a robust approach is adopted based on the H_∞ control and Lyapunov-Krasovskii theorem. In addition, it is suggested that the proposed robust control method can achieve the $\mathcal{L}_2$ string stability of the vehicle platoon under external disturbances and communication delays. To verify the effectiveness of the proposed control method, the simulation of a heterogeneous vehicle platoon running along two curved roads expressed by the cubic and 4th-degree polynomial is conducted. The simulation results reveal that the proposed technique is efficacious and outstanding for fulfilling the curved platoon control requirements compared with traditional vehicle platoon control methodology.

Appendix A
Proof of the Theorem 1

First, consider the case $h_1 \geq h_2$. Choose the Lyapunov-Krasovskii functional to be

$$V(\Xi_t) = V_1(\Xi_t) + V_2(\Xi_t) + V_3(\Xi_t) \tag{50}$$

where

$$\begin{aligned} V_1(\Xi_t) &= (\Xi(t))^T P \Xi(t) \\ V_2(\Xi_t) &= \int_{t-h_1}^{t} (\Xi(s))^T Q_1 \Xi(s) ds \\ &\quad + \int_{t-h_2}^{t} (\Xi(s))^T Q_2 \Xi(s) ds \\ V_3(\Xi_t) &= \int_{-h_1}^{0} \int_{t+\theta}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_1 \dot{\Xi}(s) ds d\theta \\ &\quad + \int_{-h_2}^{0} \int_{t+\theta}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_2 \dot{\Xi}(s) ds d\theta \\ &\quad + \int_{-h_1}^{-h_2} \int_{t+\theta}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_3 \dot{\Xi}(s) ds d\theta \end{aligned}$$

In above equations, $P > 0$, $Q_m \geq 0$, $m = 1, 2$ and $Y_l > 0$, $l = 1, 2, 3$ are to be determined.

Differentiating $V(\Xi_t)$ with respect to time along the trajectory of the closed platoon system (21), then the following equation can be got

$$\dot{V}(\Xi_t) = \dot{V}_1(\Xi_t) + \dot{V}_2(\Xi_t) + \dot{V}_3(\Xi_t) \tag{51}$$

where $\dot{V}_1(\Xi_t)$, $\dot{V}_2(\Xi_t)$ and $\dot{V}_3(\Xi_t)$ are expressed by (52) shown at the bottom of this page with

$$\begin{aligned} (\Omega(t))^T &= \Big[(\Xi(t))^T,\ (\Xi(t-\varepsilon))^T,\ (\Xi(t-\lambda))^T, \\ &\qquad (\Xi(t-h_1))^T,\ (\Xi(t-h_2))^T, (\hat{w}(t))^T\Big], \\ \tilde{\mathcal{C}} &= \begin{bmatrix} \bar{A} & \bar{B}K_1\mathcal{H}_1 & \bar{B}K_2\mathcal{H}_2 & 0 & 0 & \hat{\mathcal{G}} \end{bmatrix}, \\ \mathcal{C} &= \operatorname{diag}\left\{ Q_1 + Q_2 \quad 0 \quad 0 \quad -Q_1 \quad -Q_2 \quad -\gamma_1^2 \right\}. \end{aligned}$$

To address the integral that contains the differential in $\dot{V}_3$ and reduce the conservation of the control system, the free-weighting-matrix (FWM) approach [28] is adopted.

According to the Newton-Leibnitz formula

$$\Xi(t) - \Xi(t-h_{tt}) - \int_{t-h_{tt}}^{t} \dot{\Xi}(s) ds = 0,\ tt = 1, 2, 3, \tag{53}$$

the following equations are correct for any matrices $\mathcal{D}, \mathcal{U}$ and $\mathcal{R}$ with appropriate dimensions:

$$\begin{cases} 2(\Omega(t))^T \mathcal{D}\mathcal{I}_2\Omega(t) - 2(\Omega(t))^T \mathcal{D} \int_{t-h_1}^{t} \dot{\Xi}(s) ds = 0 \\ 2(\Omega(t))^T \mathcal{U}\mathcal{I}_3\Omega(t) - 2(\Omega(t))^T \mathcal{U} \int_{t-h_2}^{t} \dot{\Xi}(s) ds = 0 \\ 2(\Omega(t))^T \mathcal{R}\mathcal{I}_4\Omega(t) - 2(\Omega(t))^T \mathcal{R} \int_{t-h_1}^{t-h_2} \dot{\Xi}(s) ds = 0 \end{cases} \tag{54}$$

$$\dot{V}_1(\Xi_t) = 2\left(\bar{A}\Xi(t) + \bar{B}K_1\mathcal{H}_1\Xi(t-\varepsilon) + \bar{B}K_2\mathcal{H}_2\Xi(t-\lambda) + \hat{\mathcal{G}}\hat{w}\right)^T P\Xi(t) = 2(\Omega(t))^T \tilde{\mathcal{C}} P \mathcal{I}_1 \Omega(t) \tag{52a}$$

$$\begin{aligned} \dot{V}_2(\Xi_t) &= (\Xi(t))^T Q_1 \Xi(t) - (\Xi(t-h_1))^T Q_1 \Xi(t-h_1) + (\Xi(t))^T Q_2 \Xi(t) - (\Xi(t-h_2))^T Q_2 \Xi(t-h_2) \\ &= (\Omega(t))^T \mathcal{C}\Omega(t) \end{aligned} \tag{52b}$$

$$\begin{aligned} \dot{V}_3(\Xi_t) &= h_1\left(\dot{\Xi}(t)\right)^T \mathcal{Y}_1 \dot{\Xi}(t) - \int_{t-h_1}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_1 \dot{\Xi}(s) ds + h_2\left(\dot{\Xi}(t)\right)^T \mathcal{Y}_2 \dot{\Xi}(t) - \int_{t-h_2}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_2 \dot{\Xi}(s) ds \\ &\quad + k_s h_3\left(\dot{\Xi}(t)\right)^T \mathcal{Y}_3 \dot{\Xi}(t) - \int_{t-h_1}^{t-h_2} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_3 \dot{\Xi}(s) ds \\ &= (\Omega(t))^T \tilde{\mathcal{C}}^T \left(h_1\mathcal{Y}_1 + h_2\mathcal{Y}_2 + k_s h_3 \mathcal{Y}_3\right) \tilde{\mathcal{C}}\Omega(t) - \int_{t-h_1}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_1 \dot{\Xi}(s) ds \\ &\quad - \int_{t-h_2}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_2 \dot{\Xi}(s) ds - \int_{t-h_1}^{t-h_2} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_3 \dot{\Xi}(s) ds \end{aligned} \tag{52c}$$

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where

$$\mathcal{I}_2 = \begin{bmatrix} I_{2n} & 0 & 0 & -I_{2n} & 0 & 0 \end{bmatrix},$$
$$\mathcal{I}_3 = \begin{bmatrix} I_{2n} & 0 & 0 & 0 & -I_{2n} & 0 \end{bmatrix},$$
$$\mathcal{I}_4 = \begin{bmatrix} 0 & 0 & 0 & -I_{2n} & I_{2n} & 0 \end{bmatrix}.$$

On the other hand, for any matrices $\mathcal{X}_{jj} \geq 0$, $\mathcal{F}_{jj} \geq 0$, and $O_{jj} \geq 0$, $j = 1, \ldots, 6$, and any appropriately dimensioned matrices, $\mathcal{X}_{jk}$, $\mathcal{F}_{jk}$, and O_{jk}, $j < k \leq 6$, the following equation holds:

$$(\Omega(t))^T \Lambda \Omega(t) = 0 \tag{55}$$

where $\Lambda_{jk} = h_1(\mathcal{X}_{jk} - \mathcal{X}_{jk}) + h_2(\mathcal{F}_{jk} - \mathcal{F}_{jk}) + h_3(O_{jk} - O_{jk})$, $j = 1, \cdots, 6$, $j < k \leq 6$.

Adding (53) to (51), and using (54) and (55), then the following equation can be got

$$\begin{aligned} &\dot{V}(\Xi_t) + (Z(t))^T Z(t) - \gamma_1^2 (\hat{w}(t))^T \hat{w}(t) \\ &= (\Omega(t))^T \Theta_1 \Omega(t) - \int_{t-h_1}^{t} (\Pi(t,s))^T \Theta_2 \Pi(t,s) ds \\ &\quad - \int_{t-h_2}^{t} (\Pi(t,s))^T \Theta_3 \Pi(t,s) ds \\ &\quad - \int_{t-h_1}^{t-h_2} (\Pi(t,s))^T \Theta_4 \Pi(t,s) ds \end{aligned} \tag{56}$$

where $(\Pi(t,s))^T = [\,(\Omega(t))^T \ (\dot{\Xi}(t))^T\,]$,

$$\begin{aligned} \Theta_1 = &\left\{ \left(\tilde{\mathcal{C}}\right)^T P \mathcal{I}_1 + \mathcal{D}\mathcal{I}_2 + \mathcal{U}\mathcal{I}_3 + \mathcal{R}\mathcal{I}_4 \right\}_{sym} + \mathcal{C} \\ &+ \left(\tilde{\mathcal{C}}\right)^T (h_1 \mathcal{Y}_1 + h_2 \mathcal{Y}_2 + k_s h_3 \mathcal{Y}_3) \tilde{\mathcal{C}}_1 \end{aligned} \tag{57}$$

$$\Theta_2 = \begin{bmatrix} \mathcal{X} & \mathcal{D} \\ * & \mathcal{Y}_1 \end{bmatrix} \geq 0 \tag{58}$$

$$\Theta_3 = \begin{bmatrix} \mathcal{F} & \mathcal{U} \\ * & \mathcal{Y}_2 \end{bmatrix} \geq 0 \tag{59}$$

$$\Theta_4 = \begin{bmatrix} O & \mathcal{R} \\ * & \mathcal{Y}_3 \end{bmatrix} \geq 0 \tag{60}$$

Note that, in (57), $k_s = 1$. Obviously, when $\Theta_1 \leq 0$, and $\Theta_{tt} \geq 0$, $tt = 2, 3, 4$, then

$$\begin{aligned} &\int_0^\infty \left((Z(t))^T Z(t) - \gamma_1^2 (\hat{w}(t))^T \hat{w}(t) \right) dt \\ &< V(\Xi_0) - V(\Xi_\infty) \end{aligned} \tag{61}$$

where Ξ_0 and Ξ_∞ are, respectively, the states of vehicle platoon system (21) at time $t = 0$ and $t = \infty$. Under the zero initial condition, i.e., $V(\Xi_0) = 0$, and $V(\Xi_\infty) > 0$, the following inequality is obtained

$$\int_0^\infty \left((Z(t))^T Z(t) - \gamma_1^2 (\hat{w}(t))^T \hat{w}(t) \right) dt < 0 \tag{62}$$

On the other hand, it can be proved that (57)–(60) also ensure $\dot{V}(\Xi_t) < -\kappa \|\Xi(t)\|^2$ for a sufficiently small $\kappa > 0$. Hence, vehicle platoon system (21) with $\sigma_i(t) = 0$ and $\hat{w}_i(t) = 0$ is asymptotically stable. To obtain the linear matrix inequality of (57), by using Schur complement, the inequality can be rewritten as

$$\begin{bmatrix} \Phi_2 & \sqrt{h_1}\left(\tilde{\mathcal{C}}\right)^T & \sqrt{h_2}\left(\tilde{\mathcal{C}}\right)^T & \sqrt{h_3}\left(\tilde{\mathcal{C}}\right)^T & \mathcal{I}_1^T C^T \\ * & -(\mathcal{Y}_1)^{-1} & 0 & 0 & 0 \\ * & * & -(\mathcal{Y}_2)^{-1} & 0 & 0 \\ * & * & * & -(\mathcal{Y}_3)^{-1} & 0 \\ * & * & * & * & -I \end{bmatrix} \leq 0 \tag{63}$$

where

$$\Phi_2 = \Phi_1 + \{\mathcal{D}\mathcal{I}_2 + \mathcal{U}\mathcal{I}_3 + \mathcal{R}\mathcal{I}_4\}_{sym} + h_1 \mathcal{X} + h_2 \mathcal{F} + h_3 O.$$

Define $\Theta = \mathrm{diag}\{\,\Sigma \ I \ I \ I \ I\,\}$ and $\Sigma = \mathrm{diag}\{\mathcal{P}, \ \mathcal{P}, \mathcal{P}, \ \mathcal{P}, \ \mathcal{P}, \ I\}$, where $\mathcal{P} = (P)^{-1}$. Pre-multiply and post-multiply (63) by Θ, and let $\mathcal{K}_1 = K_1 \mathcal{H}_1 \mathcal{P}$, $\mathcal{K}_2 = K_2 \mathcal{H}_2 \mathcal{P}$, $(\mathcal{Y}_j)^{-1} = \varpi_j$, $j = 1, 2, 3$, $\tilde{\mathcal{D}} = \Sigma \mathcal{D} \mathcal{P}$, $\tilde{\mathcal{U}} = \Sigma \mathcal{U} \mathcal{P}$, $\tilde{\mathcal{R}} = \Sigma \mathcal{R} \mathcal{P}$, $\tilde{\mathcal{X}} = \Sigma \mathcal{X} \Sigma$, $\tilde{\mathcal{F}} = \Sigma \mathcal{F} \Sigma$, $\tilde{\mathcal{O}} = \Sigma O \Sigma$, and $\tilde{\mathcal{Q}}_k = \mathcal{P} Q_k \mathcal{P}$, $k = 1, 2$, then (35) is obtained.

In addition, pre-multiply and post-multiply (58)–(60) by $\mathrm{diag}\{\Sigma, \ \mathcal{P}\}$, (64)–(66) can be got.

$$\begin{bmatrix} \tilde{\mathcal{X}} & \tilde{\mathcal{D}} \\ * & \mathcal{P}\mathcal{Y}_1\mathcal{P} \end{bmatrix} > 0 \tag{64}$$

$$\begin{bmatrix} \tilde{\mathcal{F}} & \tilde{\mathcal{U}} \\ * & \mathcal{P}\mathcal{Y}_2\mathcal{P} \end{bmatrix} > 0 \tag{65}$$

$$\begin{bmatrix} \tilde{\mathcal{O}} & \tilde{\mathcal{R}} \\ * & \mathcal{P}\mathcal{Y}_3\mathcal{P} \end{bmatrix} > 0 \tag{66}$$

Then, by using Lemma 1, (36)–(38) is obtained.

In addition, when $h_1 < h_2$, the Lyapunov-Krasovskii functional candidate is

$$V(\Xi_t) = V_1(\Xi_t) + V_2(\Xi_t) + V_4(\Xi_t) \tag{67}$$

where

$$\begin{aligned} V_4(\Xi_t) = &\int_{-h_1}^{0} \int_{t+\theta}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_1 \dot{\Xi}(s) ds d\theta \\ &+ \int_{-h_2}^{0} \int_{t+\theta}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_2 \dot{\Xi}(s) ds d\theta \\ &+ \int_{-h_2}^{-h_1} \int_{t+\theta}^{t} \left(\dot{\Xi}(s)\right)^T \mathcal{Y}_3 \dot{\Xi}(s) ds d\theta \end{aligned}$$

(54) can be reformulated as

$$\begin{cases} 2(\Omega(t))^T \mathcal{D}\mathcal{I}_2 \Omega(t) - 2(\Omega(t))^T \mathcal{D} \int_{t-h_1}^{t} \dot{\Xi}(s) ds = 0 \\ 2(\Omega(t))^T \mathcal{U}\mathcal{I}_3 \Omega(t) - 2(\Omega(t))^T \mathcal{U} \int_{t-h_2}^{t} \dot{\Xi}(s) ds = 0 \\ 2(\Omega(t))^T \mathcal{R}\mathcal{I}_4 \Omega(t) - 2(\Omega(t))^T \mathcal{R} \int_{t-h_2}^{t-h_1} \dot{\Xi}(s) ds = 0 \end{cases} \tag{68}$$

Then, following the procedure for the case $h_1 \geq h_2$ generate a similar result; but note that, in this case, $k_s = -1$.

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